



LOK JAGRUTI KENDRA UNIVERSITY
Integrated MSC AI/ML Div Semester – 7
Mid Semester Practical Examination - 2026

Subject Code: 150020701

Subject Name : Applied Statistics

Total Marks: 30

Date: 27/09/2026

SET-1

1	The daily exercise time (in minutes) and corresponding calories burned by 7 people are given below. Write this data into a CSV file named exercise_data.csv with columns Exercise_Minutes and Calories_Burned. <table><tr><th>Exercise_Minutes</th><th>Calories_Burned</th></tr><tr><td>15</td><td>80</td></tr><tr><td>20</td><td>110</td></tr><tr><td>30</td><td>150</td></tr><tr><td>40</td><td>190</td></tr><tr><td>50</td><td>240</td></tr><tr><td>60</td><td>300</td></tr><tr><td>75</td><td>380</td></tr></table> Perform the following tasks: a) Create the CSV file exercise_data.csv using the given data. (2) b) Assign a Weight of 2 to people with Exercise_Minutes greater than 50, and 1 otherwise, then compute the weighted mean of Calories_Burned using these weights. (3) c) Calculate the mean, median, mode, and 10% trimmed mean of Exercise_Minutes. (4) c) Calculate the variance, standard deviation, and IQR of Exercise_Minutes. (3)	Exercise_Minutes	Calories_Burned	15	80	20	110	30	150	40	190	50	240	60	300	75	380	12
Exercise_Minutes	Calories_Burned																	
15	80																	
20	110																	
30	150																	
40	190																	
50	240																	
60	300																	
75	380																	
2	Generate random samples from Bernoulli, Binomial, and Poisson distributions using NumPy/SciPy.	13																
3	Viva	05																



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SET-2

1	The monthly electricity consumption of 10 households is given below. Create a CSV file named electricity_data.csv with the columns House, Family_Members, and Electricity_Units. <table><tr><th>House</th><th>Family_Members</th><th>Electricity_Units</th></tr><tr><td>H1</td><td>2</td><td>120</td></tr><tr><td>H2</td><td>3</td><td>150</td></tr><tr><td>H3</td><td>4</td><td>180</td></tr><tr><td>H4</td><td>4</td><td>175</td></tr><tr><td>H5</td><td>5</td><td>210</td></tr><tr><td>H6</td><td>3</td><td>160</td></tr><tr><td>H7</td><td>6</td><td>250</td></tr><tr><td>H8</td><td>5</td><td>220</td></tr><tr><td>H9</td><td>4</td><td>190</td></tr><tr><td>H10</td><td>7</td><td>300</td></tr></table> Perform the following tasks: a) Create the CSV file electricity_data.csv using the given data and load it into a pandas DataFrame. (2) b) Calculate the mean, median, mode, and 10% trimmed mean of Electricity_Units. (4) c) Calculate the variance, standard deviation, and IQR of Electricity_Units. (3) d) Calculate the coefficient of variation (CV) and identify any outliers using the IQR method in Electricity_Units. (3)	House	Family_Members	Electricity_Units	H1	2	120	H2	3	150	H3	4	180	H4	4	175	H5	5	210	H6	3	160	H7	6	250	H8	5	220	H9	4	190	H10	7	300	12
House	Family_Members	Electricity_Units																																	
H1	2	120																																	
H2	3	150																																	
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H4	4	175																																	
H5	5	210																																	
H6	3	160																																	
H7	6	250																																	
H8	5	220																																	
H9	4	190																																	
H10	7	300																																	
2	Simulate a sample space for two dice in Python, enumerate events, (2) Compute Probabilities: a) sum of the two dice is even. (2) b) at least one die shows 6. (2) c) sum of the dice is greater than 8. (2) d) Calculate basic set operations (Union, Intersection and Complement). (5)	13																																	
3	Viva	05																																	